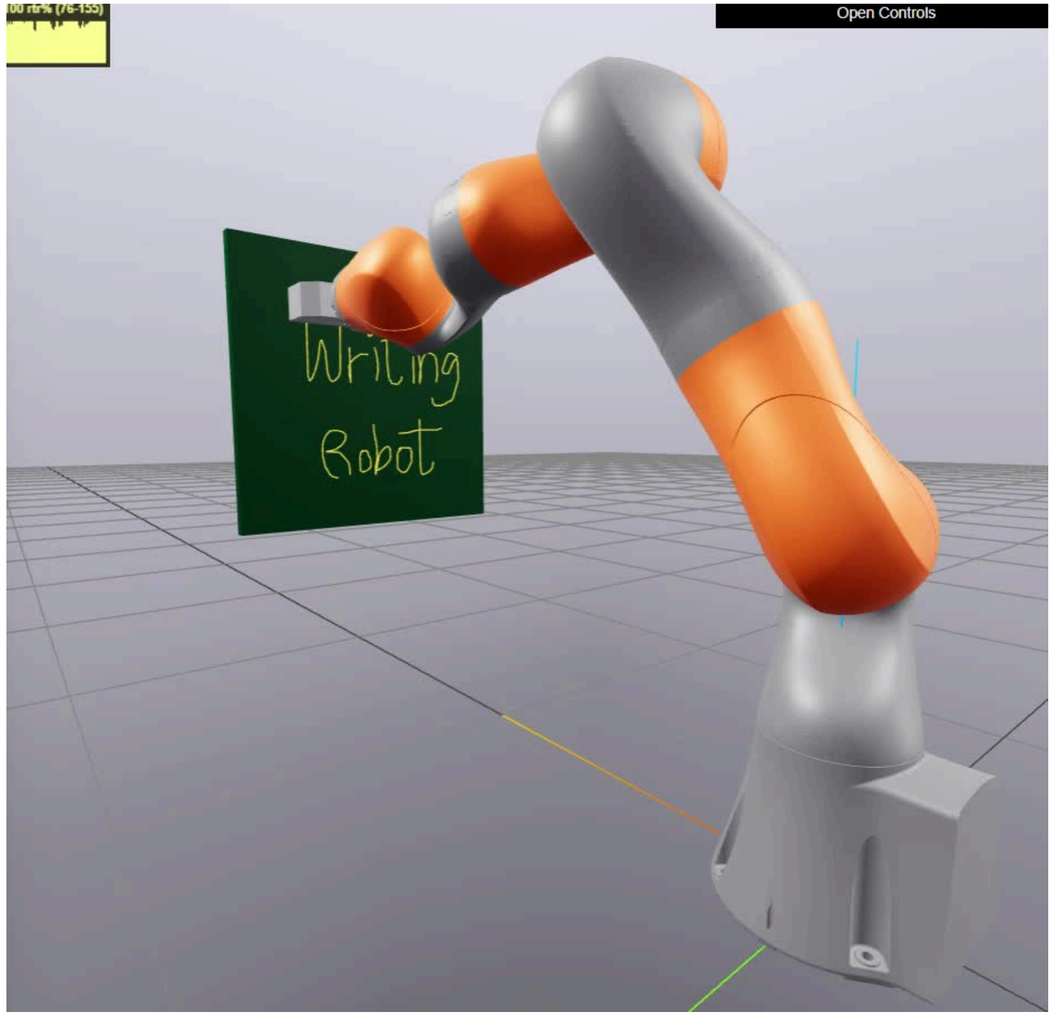
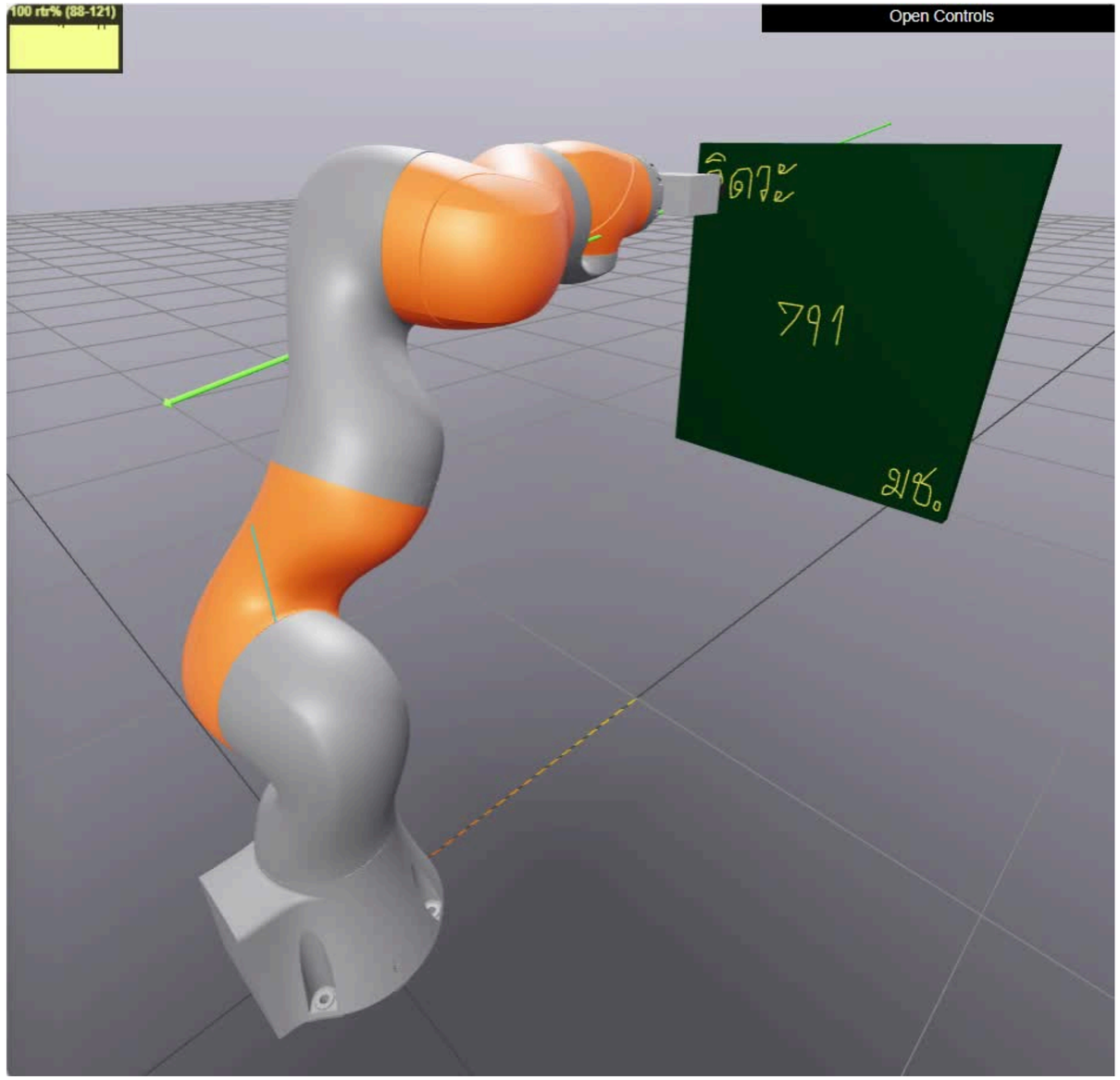
WRITING ROBOT

Anurak Narak 690631040



Writing Robot in Simulation



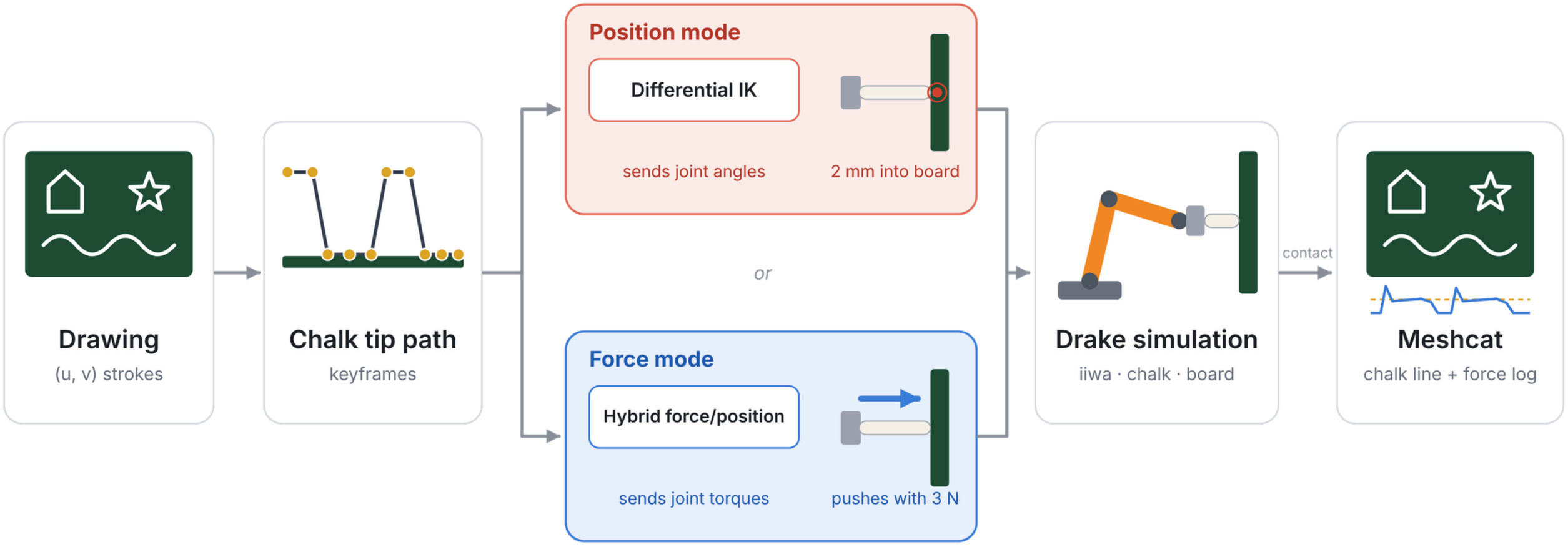
Running the Simulation

1. Open Chalk Drawing Board.html

2. Write anything on the Board

3. Copy JSON Format in to Heading 4 from the code

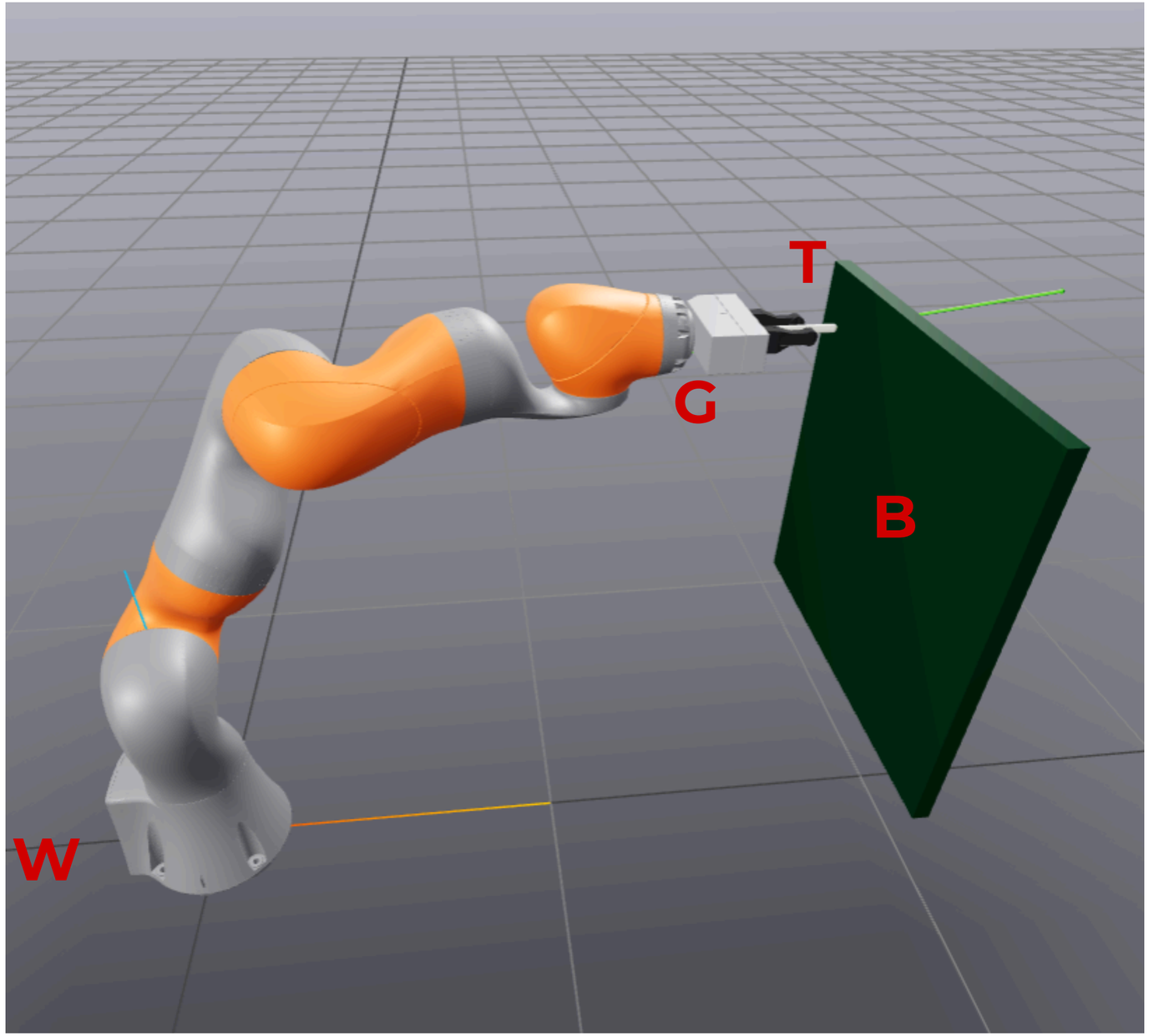
How the System Works



Building the Scene

Model	Source	Welded to
KUKA iiwa (7 joints)	Drake model library	World
WSG gripper	manipulation package	iiwa link 7
Chalk	My SDF file	Gripper
Board (50 × 50 cm)	My SDF file	World, at X_WB

Frames and Poses in the Simulation



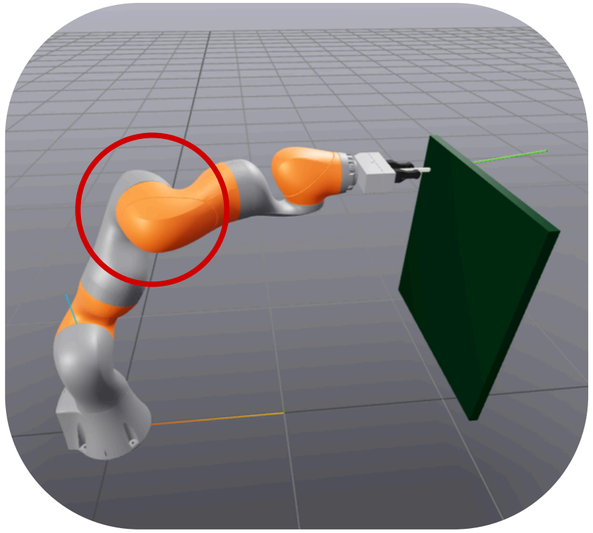
X_WB → Board in the world

X_GT → Chalk tip relative to the gripper

X_WT → Chalk tip in the world

X_WG → Gripper in the world

Finding the Start Posture with IK Optimization



$$q^\star \;=\; \arg\min_{q \in \mathbb{R}^7} \quad \left\| \, q - q_{\text{comfort}} \, \right\|_2^2$$

$$\text{subject to} \quad p_{WT}(q) \;=\; X_{WB} \begin{bmatrix} 0 \\ 0 \\ h \end{bmatrix} \qquad \text{tip position (3 constraints)}$$

$$\angle\big(\, z_T(q), \; -n_W \, \big) \;\leq\; 10^{-3} \text{ rad} \qquad \text{chalk} \perp \text{board (2 constraints)}$$

$$q_{\min} \;\leq\; q \;\leq\; q_{\max} \qquad \text{joint limits}$$

Solving and selection (nonconvex → multi-start, solver: SNOPT)

Solve from 20 initial guesses, then keep the solution closest to q_{comfort} with

$$\text{margin}(q^\star) \;=\; \min_i \, \min \left(\, q_i^\star - q_{i,\min}, \;\; q_{i,\max} - q_i^\star \, \right) \;\geq\; 0.3 \text{ rad}$$

Turning a Drawing into Robot Motion

Position Control with Differential IK

Chalk tip trajectory → Gripper trajectory → Differential IK → Joint angles → iiwa (stiff PID)

How the chalk touches the board

The tip is commanded 2 mm into the board

The joints are very stiff → the robot pushes hard

Position Control with Differential IK

$$\dot{q}^\star,\ \alpha^\star \;=\; \underset{\dot{q}\in\mathbb{R}^7,\ \alpha}{\arg\min} \quad -100\,\alpha \;+\; \left\| P\left(\dot{q} - K\left(q_{\text{write}} - q\right)\right)\right\|_2^2$$

$$\text{subject to} \quad J(q)\,\dot{q} \;=\; \alpha\, V_{\text{desired}} \qquad \text{follow the gripper path}$$

$$0 \;\le\; \alpha \;\le\; 1 \qquad \text{slow down if needed}$$

$$\dot{q}_{\min} \;\le\; \dot{q} \;\le\; \dot{q}_{\max} \qquad \text{joint velocity limits}$$

Every time step (Drake DifferentialInverseKinematicsIntegrator)

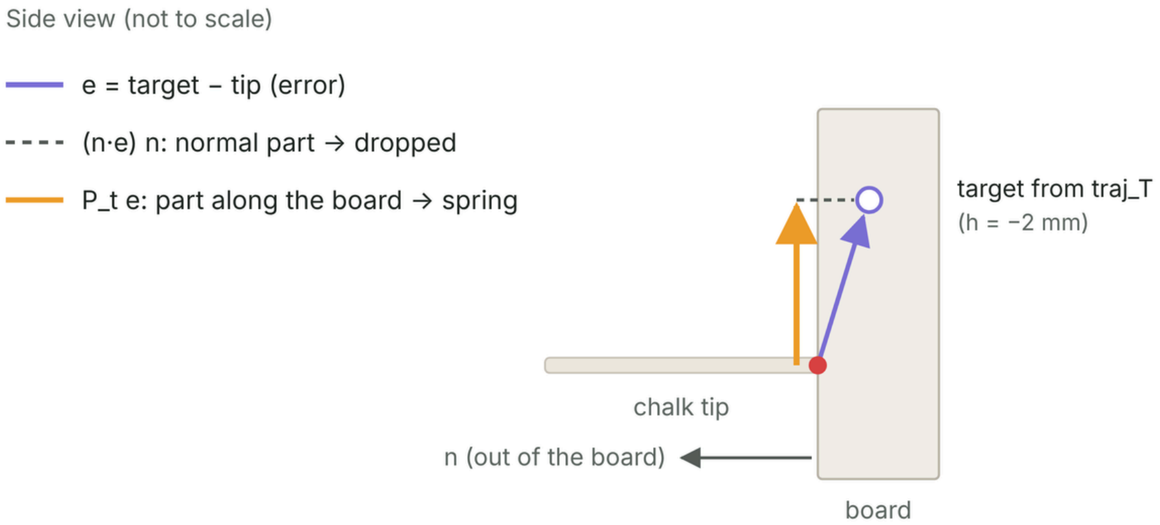
$$q_{\text{cmd}} \;\leftarrow\; q_{\text{cmd}} + \Delta t\,\dot{q}^\star \quad\longrightarrow\quad \text{iiwa position input} \;\longrightarrow\; \text{stiff joint PID}$$

Hybrid Force/Position Control

Control position along the board, control force into the board

Direction	What we control	How
Along the board	Position	Virtual spring + damper
Into the board	Force	Push with F_DESIRED = 3 N
Rotation	Orientation	Rotational spring + damper

Hybrid Force/Position Control



Manipulator Equations

Chp_8 $$M(q)\ddot{q} + C(q,\dot{q})\dot{q} = \tau_g(q) + u + \sum_i J_i^T(q) f_{c_i}$$

1 Virtual spring-damper $$a_{\text{rot}} = K_r\, e_R + D_r\,(\omega_d - \omega), \qquad a_{\text{lin}} = K_p\,(p_d - p) + K_d\,(v_d - v)$$

2 Pen down: drop the normal part $$a_{\text{lin}} \leftarrow P_t\, a_{\text{lin}}, \qquad P_t = I - n\,n^T$$

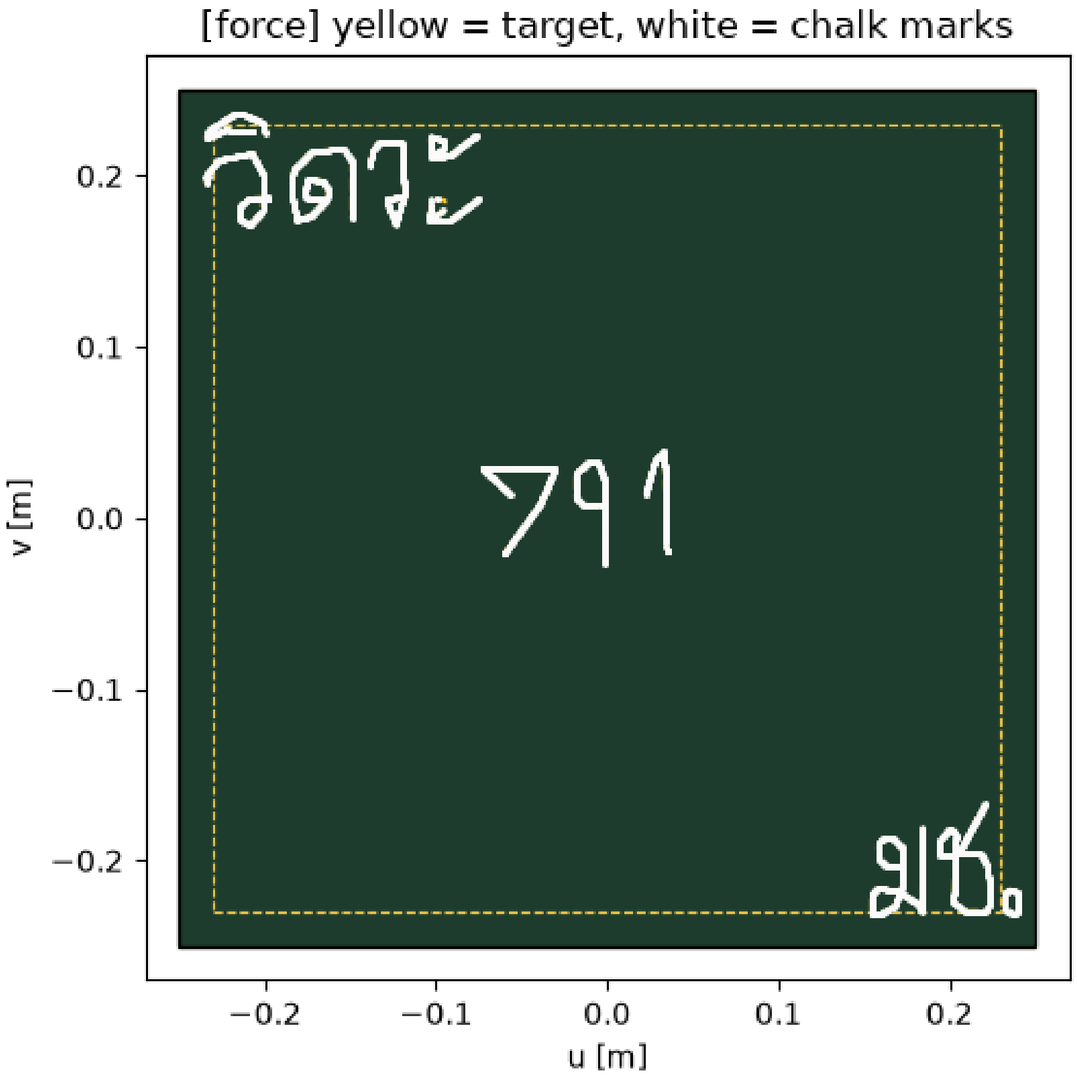
3 Acceleration → force $$F = \begin{bmatrix} F_{\text{rot}} \\ F_{\text{lin}} \end{bmatrix} = \Lambda \begin{bmatrix} a_{\text{rot}} \\ a_{\text{lin}} \end{bmatrix}, \qquad \Lambda = \left(J\,M^{-1}J^T\right)^{-1}$$ Lambda: operational space inertia
M: mass matrix

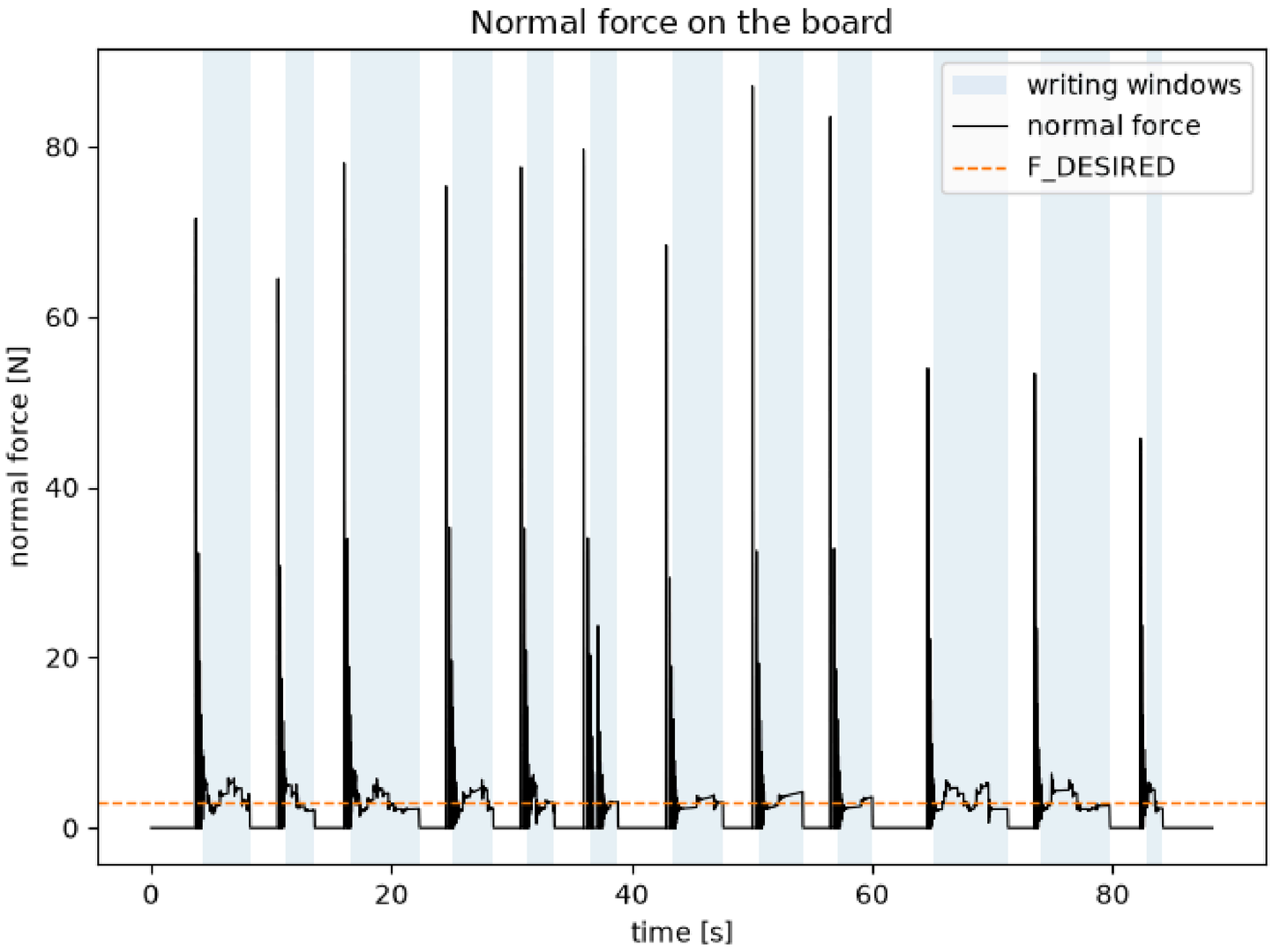
4 Pen down: push into the board $$F_{\text{lin}} \leftarrow F_{\text{lin}} - s\,F_{\text{DESIRED}}\,n - D_n\,(n^T v)\,n$$ Dn: damping

5 Joint torques + posture $$\tau = \underline{J^T F} + \underline{N\,M\left(K_q\,(q_{\text{write}} - q) - D_q\,\dot{q}\right)}, \qquad N = I - J^T \Lambda\, J\, M^{-1}$$ N: null space

task-space (operational space) control + **joint-space posture control in the null space**

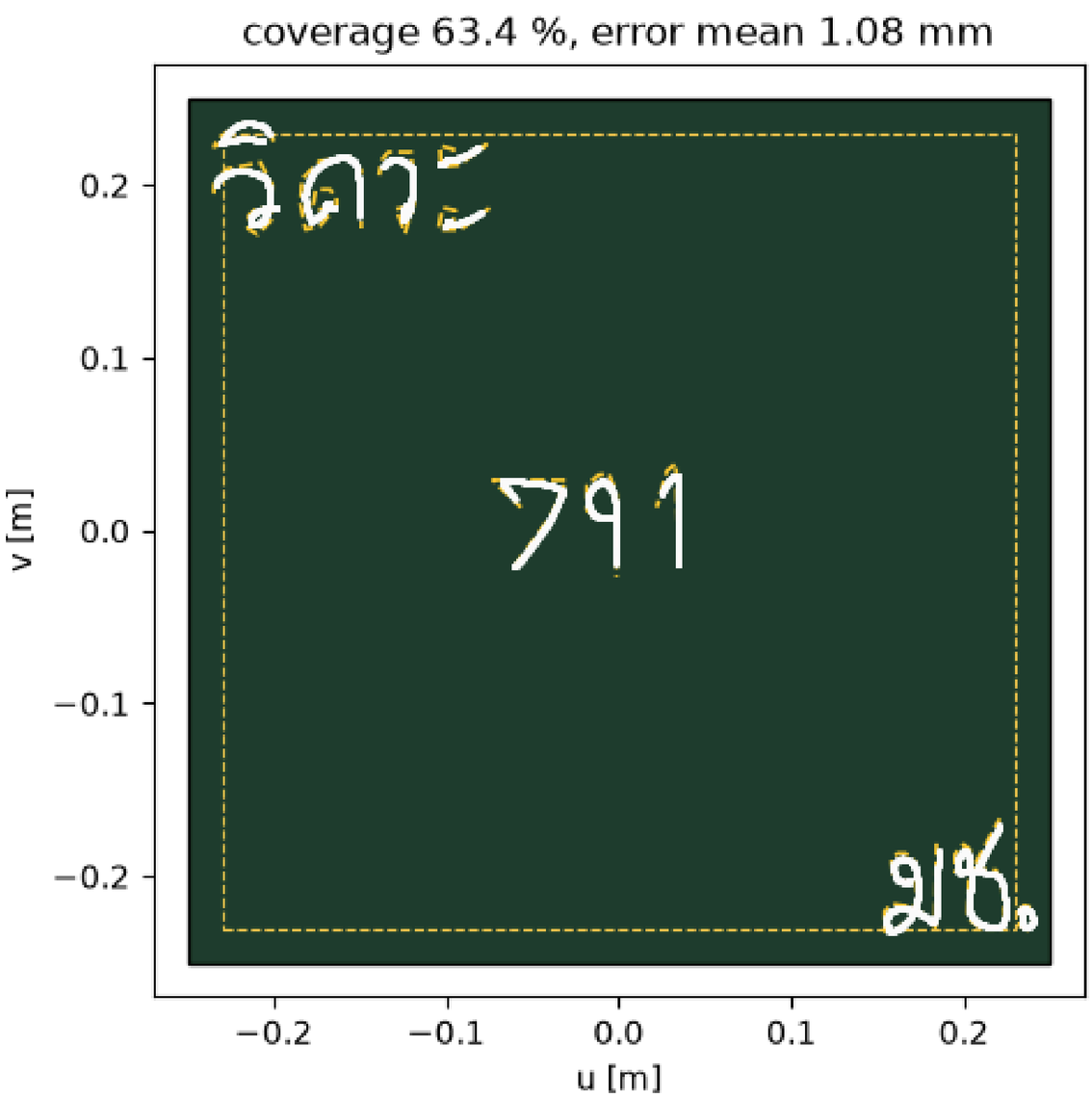
Evaluation: Force Control



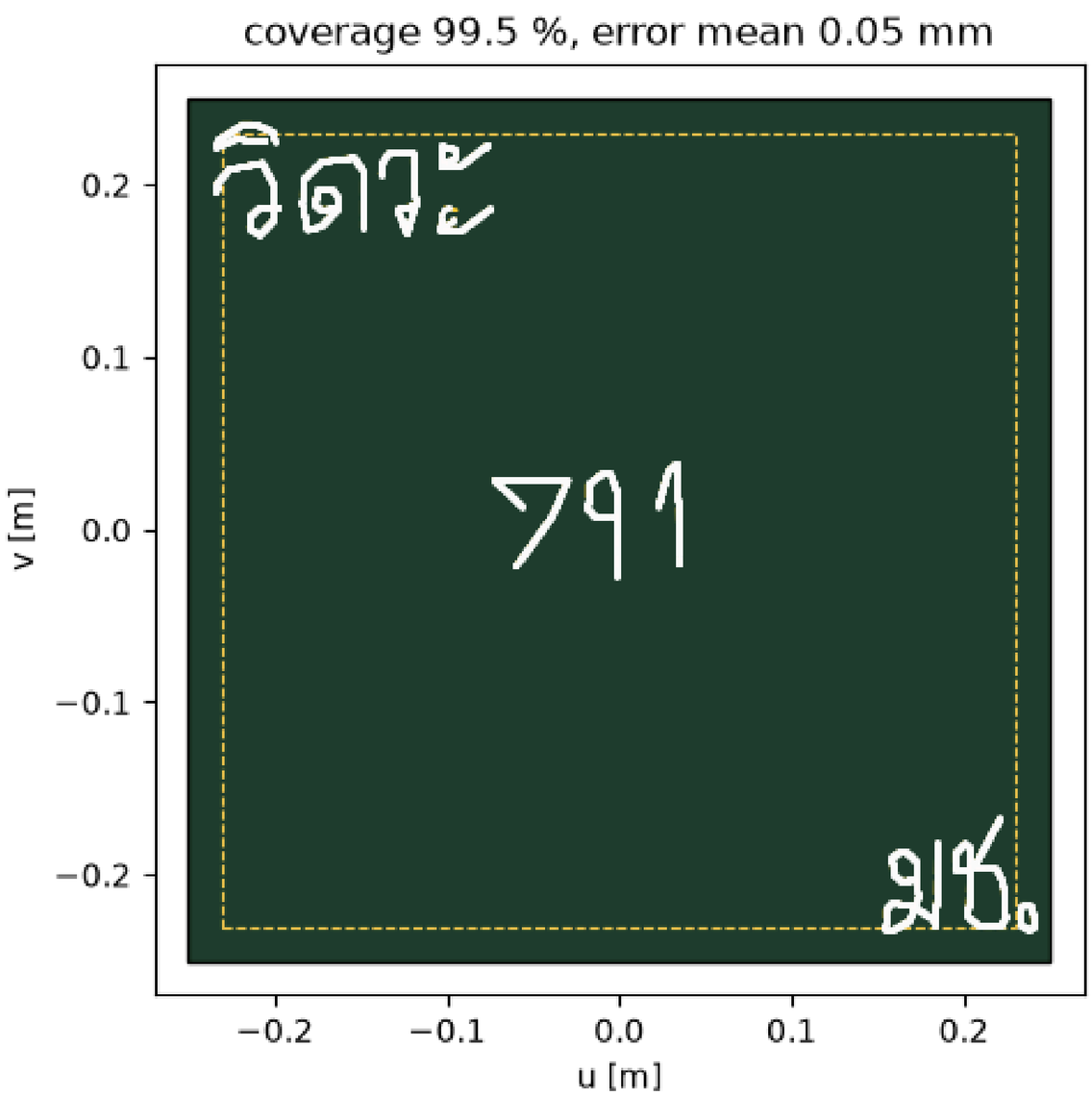


Evaluation: Position vs Force Control

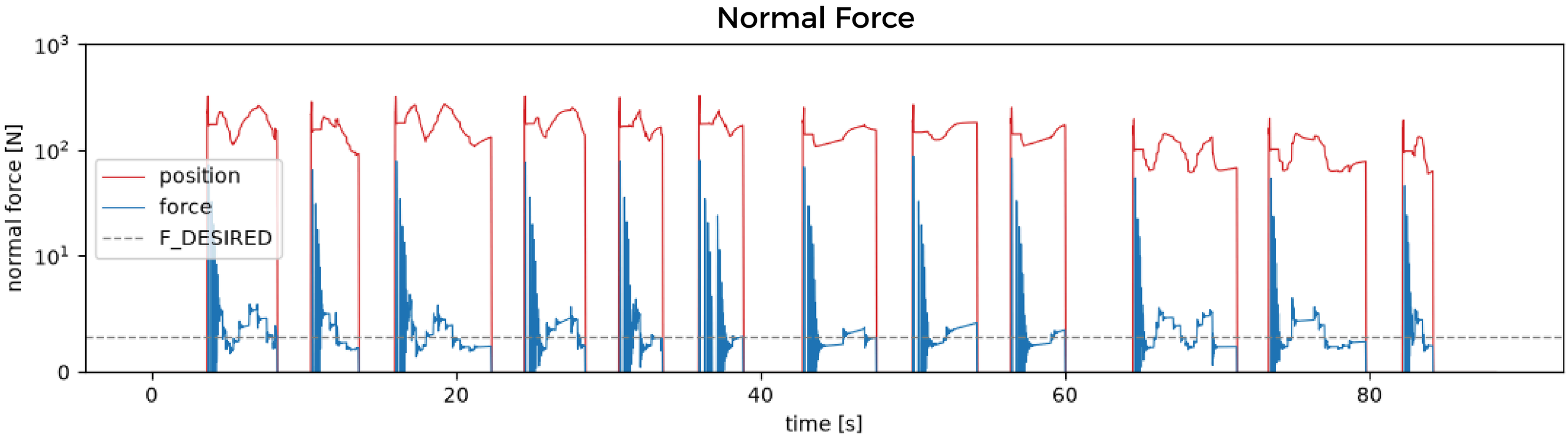
Position



Force



Evaluation: Position vs Force Control



Limitations

Open-loop force control

A short force peak when the chalk first touches the board

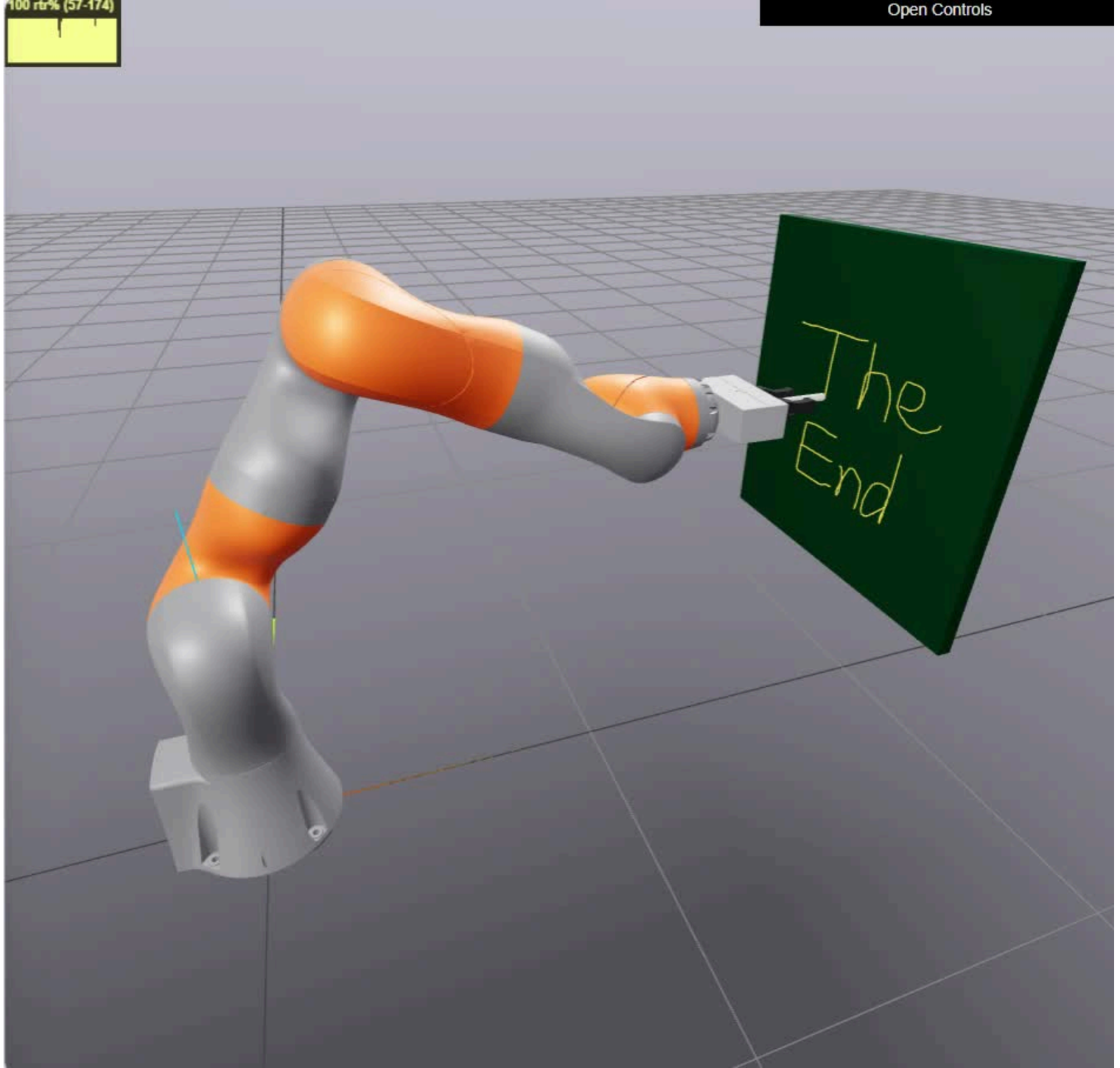
The chalk is welded to the gripper (no grasping)

Future work

Force feedback from the measured joint torques

Estimate the board pose with a camera

Calibrate the chalk tip by touching the board


Open Controls
The
End